

Interface

Robot Magazine Handling for AES/AMS

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Change history

Rev.	Release	Author	Date	Changes
1.0		Bernd Autenrieth (ASYS)	15/06/2016	Initial
1.1		Bernd Autenrieth (ASYS)	30/06/2016	2.2.4 Extension by l_lane. A separate command is sent for each magazine lane. Magazine empty information. Extension SMEMA Advanced. Sequence diagrams for magazine handling
1.2		Bernd Autenrieth (ASYS)	25/07/2016	2.2.4 value2= number of parts extended Page 8 Detail Handover completed only after arrival. 2.2.5.1.1 BOARD_DELIVERY = 0 accepted in sequence. 2.2.3 extended by interface signals.
1.3		Thomas Lang (ASYS)	21/10/2016	2.2.4 Description added 2.2.5.1.1 BOARD_RECEIVAL = 0 accepted in sequence 2.2.5.2.1 BOARD_RECEIVAL = 0, BOARD_DELIVERY = 0 accepted in sequence 3.1 Appendix State diagram added 3.2 Appendix Sequence program added
1.4		Bernd Autenrieth (ASYS)	19/12/2016	Typos fixed
1.5		Thomas Lang (ASYS)	26/01/2017	1.1 Communication default port changed 3.2 Appendix Sequence diagram modified
1.6	Approved by	Thomas Lang (ASYS)	05/03/2018	2.2.4 Magazine warning parameter s_value1 changes, applies to, e.g., AES05. More detailed description, added example.
1.7		???	???	2.2.5 Send messages cyclically until transport starts
1.8		Martin Essig (ASYS)	11/01/2019	2.2.6 BOARD_REQUIREMENT Message supplemented by parameters MagCode, MatType, and Charge
1.9		Martin Essig (ASYS)	11/02/2019	2.2.6 BOARD_REQUIREMENT Message supplemented parameter Product Name
2.0		Thomas Lang (ASYS)	17/05/2019	2.2.6 Supplements
2.1		Martin Essig (ASYS)	02.09.2020	2.2.4 Supplements „Warning“ message
2.2		Martin Essig (ASYS)	14.04.2021	2.2.4 Supplements „Warning“ message (l_remaining)
2.3		Martin Essig (ASYS)	15.06.2021	2.2.4 Supplements „Warning“ message (l_MagLevel)
2.4		Timo Meißner (ASYS)	01.06.2022	2.2.4 Supplements “Warning” message (l_magazineWidth)

Version: The version of the specification.

Release: The release date of the version – empty if the version has not been released

Author: If there are multiple authors (see cover sheet), the authors are listed here who have added version changes to the specification. If there is only one author, this field can be left empty.

Date: Change date

Change: Brief description of all the changes in a version

Abbreviations

ASYS	ASYS Automatisierungssysteme GmbH
IP	Internet Protocol
TCP	Transmission Control Protocol
XML	Extensible Markup Language
PCB	PCB

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1 Introduction

This document specifies the interface between ASYS and a robot system.

1.1 Communication

Communication between the systems is via a socket connection based on the TCP/IP protocol. Data is exchanged using messages in XML notation. The communication port must be freely configurable (default is 5000).

1.1.1 Establishing a connection

The connection between ASYS and the robot is always established by the robot. ASYS is therefore the socket server and the robot is the socket client.

1.1.2 Disconnecting

If the connection is interrupted, the robot must attempt to reestablish the connection in the background. If there is no connection between ASYS and the robot, this must be shown visually both at ASYS and at the robot.

1.2 Potential changes to specification

ASYs reserves the right to make minor changes to the specification.

2 Communication

2.1 Basic information

Communication direction:

E → A External system sends Asys
 A → E Asys sends to external system

2.2 Cyclical verification of the connection

The robot sends a keep-alive message cyclically once per second, to which ASYS must respond. If this response from Asys fails three times in a row, the robot will close and rebuild the socket (connect).

If Asys receives no keep-alive message from the robot for at least 3 seconds, ASYS (server) will close the socket and wait for a new connection (listen)

2.2.1 Keep-alive

The robot sends a keep-alive message at least once per second.

ALIVE_REQ		E → A
<pre><ASYS> <ALIVE_REQ s_source="ROBOT"/> </ASYS></pre>		
s_source	Robot name.	

2.2.2 Response of ASYS system

The machine responds to each keep-alive request from the robot.

ALIVE_RES		A → E
<pre><ASYS> <ALIVE_RES s_source="AES03D"/> </ASYS></pre>		
s_source	Name or type of the system.	

2.2.3 Request magazine status and interface signals.

With this message, the robot requests the current magazine status and the current interface signals from the system. In addition, the robot sends its current interface signals after receiving this message. (Required only after a new connection has been established. Status is sent event-driven):

REQUEST_STATEINFO	E → A
<pre><ASYS> <REQUEST_STATEINFO/> </ASYS></pre>	

2.2.4 Magazine warning reached.

A warning can be defined via the machine (number of PCBs reached in magazine). With this message, the machine announces that a magazine change will take place soon. The message is sent cyclically (cycle time default: 10 s, configurable) and upon a state change.

Any warnings (*l_come*=1) are sent only in Automatic mode. If the machine is stopped (stop/error), the warning is terminated (*l_come*=0).

A distinction has to be made between machines with buffer capacity per lane (intake/output of several magazines) and machines where only 1 magazine can be transported per lane. *L_value1* is for additional Information, essential for the current status is always *l_value2*!

WARNING		A → E
<pre><ASYS> <WARNING l_warningNb="21" s_warningText="Magazine Pre Limit reached" l_come="1" l_lane="1" l_value1="1" l_value2="0" s_MagCode="123abcDEF" l_remaining="1" l_MagLevel="0"/> </ASYS></pre>		
<i>l_warningNb</i>	Warning number.	
<i>s_warningText</i>	The text of the warning.	
<i>l_come</i>	Warning occurs / is finished: 1 = occurs 0 = finished	
<i>l_lane</i>	Lane of the magazine. A separate message is sent for each lane.	
<i>l_value1</i>	Pre warning, available machine capacity (on machines with buffer places). 0=Buffer capacity available, 1= Warning, buffer section full	
<i>l_value2</i>	Warning, remaining number of parts (0 = magazine full/empty = no remaining parts, -1 no magazine available, -2 number of parts is unknown.)	
<i>s_MagCode</i>	Magazine barcode [string (80)] from the next magazine to be dispensed (optional)	
<i>l_remaining</i>	- For tracks that take containers/magazines: Number of free buffer positions - For tracks that deliver containers/magazines: Number of containers/magazines ready for delivery (optional)	
<i>l_MagLevel</i>	Actual filling level of containers/magazines (optional)	
<i>l_magazineWidth</i>	Outside width of the magazine in tenths of a millimeter (optional)	

Examples:

```
<ASYS>
  <WARNING l_warningNb="1110" s_warningText="Magazine empty"
l_come="1" l_lane="1" l_value1="0" l_value2="0" s_MagCode=""
l_remaining="1" l_MagLevel="0" l_magazineWidth="0"/>
</ASYS>
```

- ➔ AES: Empty magazine ready for pick-up. No warning, therefore still capacity available for further empty magazines on the removal conveyor.

```
<ASYS>  
  <WARNING l_warningNb="1110" s_warningText="Magazine empty"  
    l_come="1" l_lane="1" l_value1="1" l_value2="0" s_MagCode=""  
    l_remaining="1" l_MagLevel="0" l_magazineWidth="0"/>  
</ASYS>
```

- ➔ AES: Empty magazine ready for pick-up. Warning active, therefore no capacity available for further empty magazines on the removal conveyor. This means that the machine is close to a standstill and the magazine should be picked up as soon as possible.

```
<ASYS>  
  <WARNING l_warningNb="1104" s_warningText="No magazine"  
    l_come="1" l_lane="1" l_value1="1" l_value2="-1" s_MagCode=""  
    l_remaining="1" l_MagLevel="0" l_magazineWidth="0"/>  
</ASYS>
```

- ➔ AES/AMS: Machine requires a new magazine. Warning active, buffer section is already full. This means that a new magazine can be added, but the machine already has enough magazines on the feed section.

Definition of warning numbers:

21:	No warning pending (at l_come=0)
1001...1004:	Magazine 1...4 full
1005...1008:	Magazine 1...4 empty
1104:	No magazine (-> magazine missing)
1110:	Magazine empty
1111:	Magazine full
2086:	Remaining run time, remaining capacity, warning limit reached (relating to parts in the active magazine)

2.2.5 Magazine handling via ADVANCED SMEMA over TCP/IP

No extended SMEMA interface is used at the TCP/IP XML protocol level (no hardware signals at the IO level) to change a magazine.

Cyclic sending of BOARD_PERMISSION.

- If the robot is in front of the system, the BOARD_PERMISSION telegram must be sent cyclically until transfer starts

Cyclic sending of BOARD_REQUIREMENT

- If the robot is in front of the system, the BOARD_REQUIREMENT telegram must be sent cyclically until transfer starts

Cyclic sending of BOARD_DELIVERY.

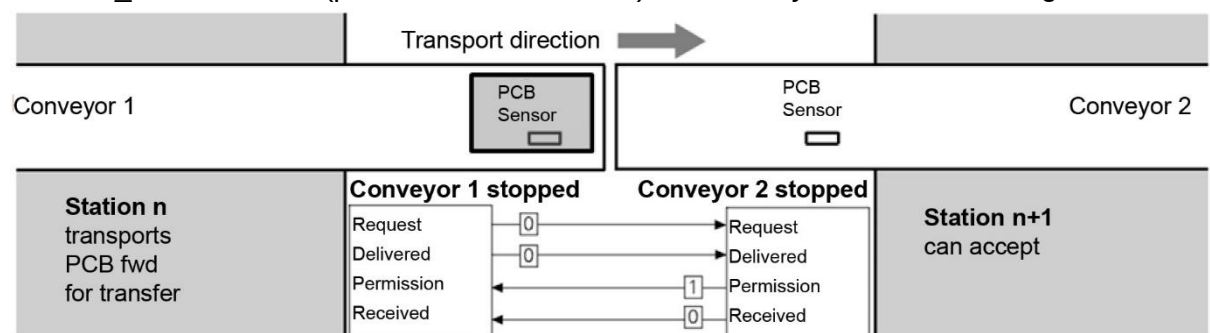
- If the robot is in front of the system, the BOARD_DELIVERY telegram must be sent cyclically until transfer starts

Cyclic sending of BOARD_RECEIVAL

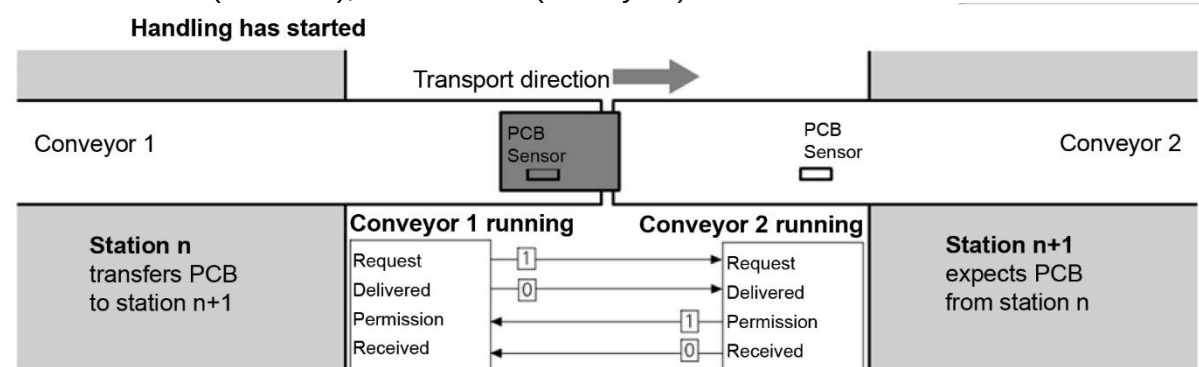
- If the robot is in front of the system, the BOARD_RECEIVAL telegram must be sent cyclically until transfer starts

2.2.5.1 Magazine handover from the machine to the robot

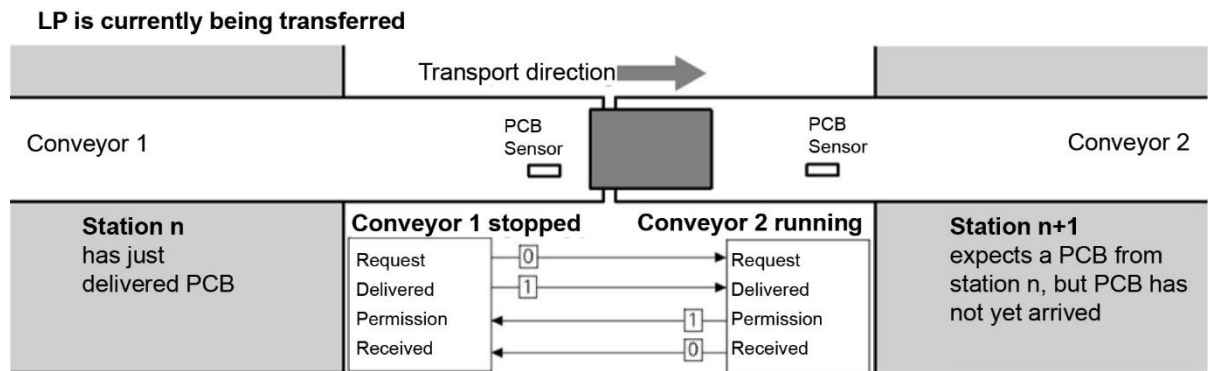
The robot is located here in front of the respective magazine and sends the telegram BOARD_PERMISSION (permission to take over) when ready to take over a magazine.



If the machine is ready, it responds with the telegram BOARD_REQUIREMENT (request). If both are active (available), the handover (conveyors) starts.

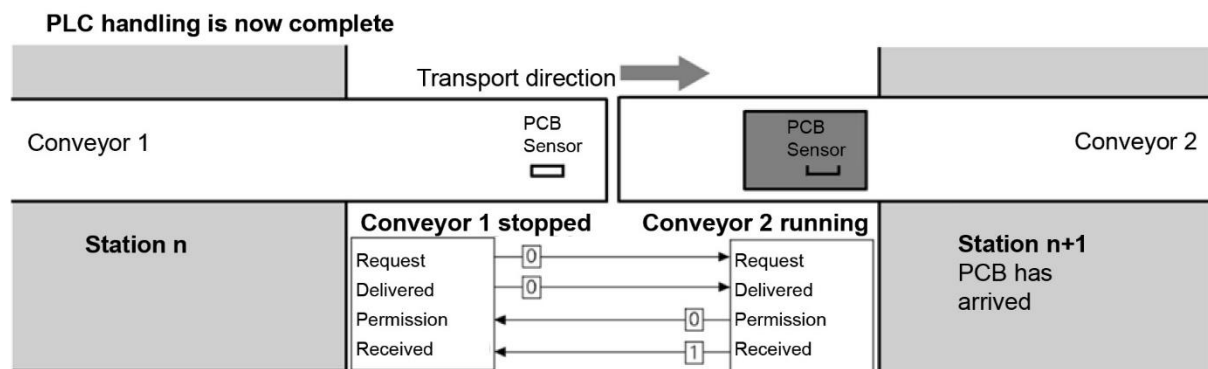


When the magazine has left the sensor of the delivering machine, it sends the message BOARD_DELIVERY (delivered -> active l_value=1) and resets the request to signal that the magazine has left the sensor.



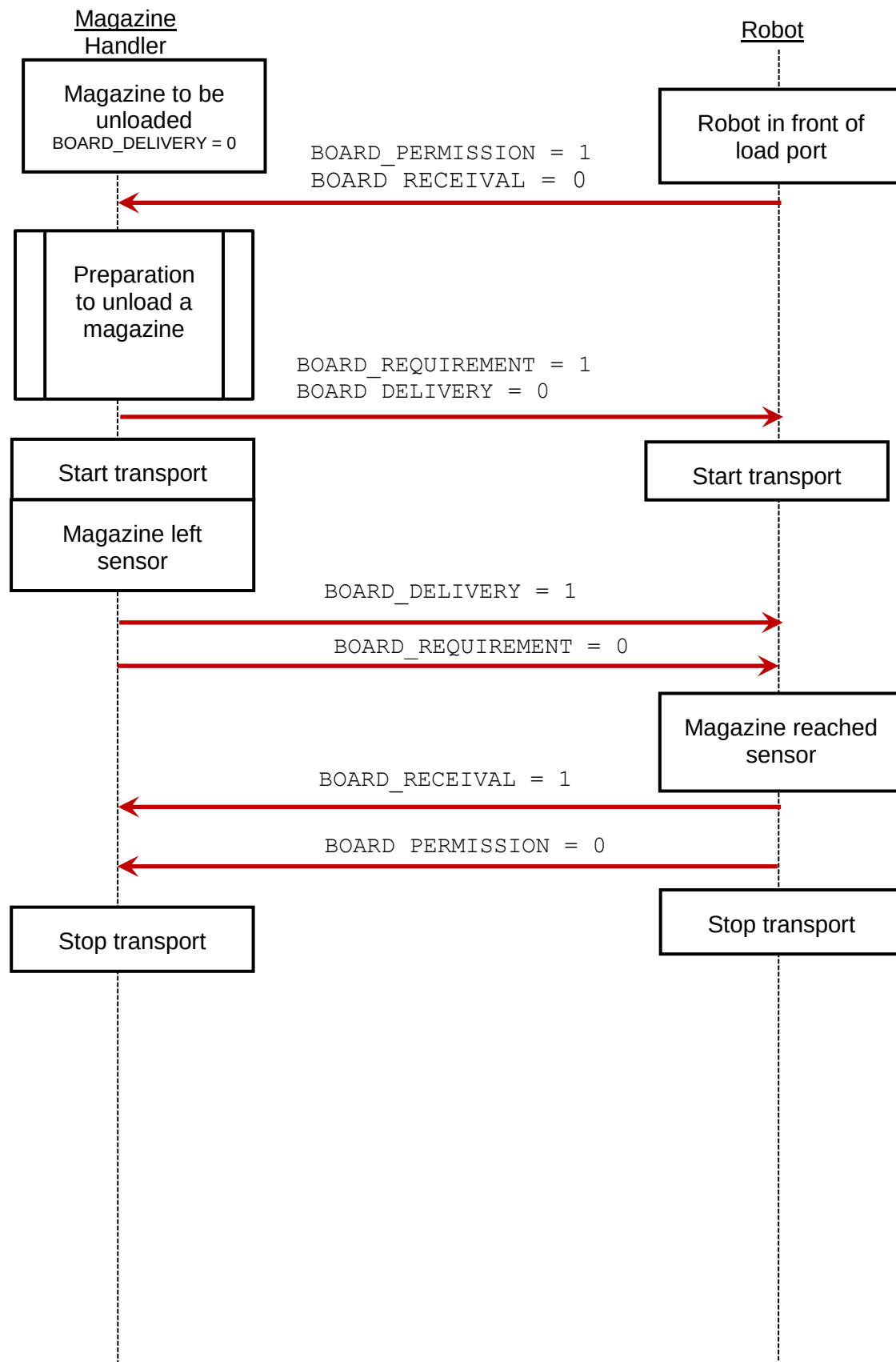
If an emergency stop is triggered in this state, the station (emergency stop active) takes away both signals. Both stations then stop the conveyors.

When the magazine has arrived at the “station n+1”, the “station n+1” sends the message BOARD_RECEIVAL (arrived -> active I_value=1) and takes away the permission (BOARD_PERMISSION -> active I_value=0).



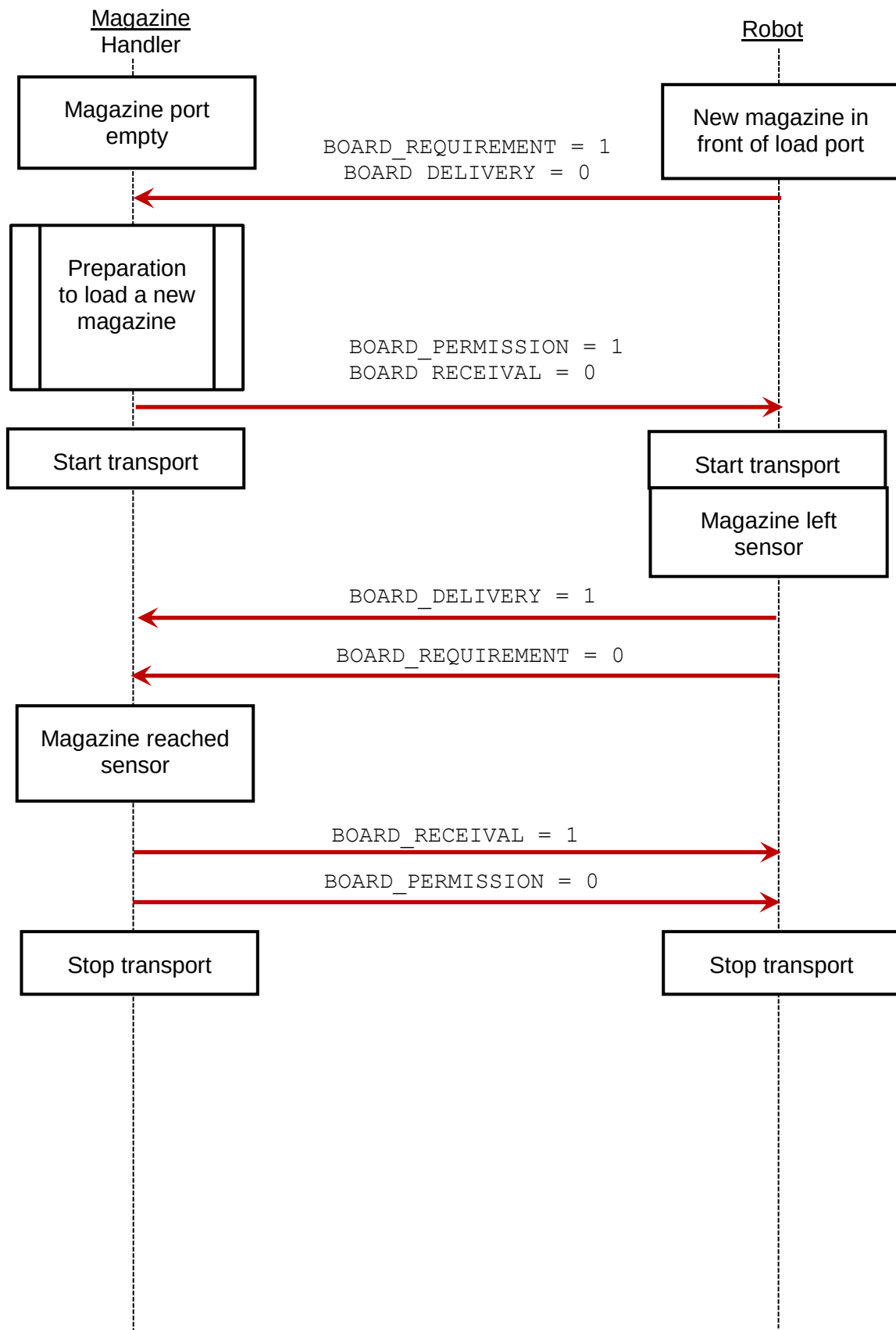
At the handover from the machine to the robot, the machine may take over new magazines only after “arrived at robot” has been set. Otherwise, there could be an involuntary handover of a second magazine after an emergency stop on the robot that is triggered after the magazine has already left the machine.

2.2.5.1.1 Sequence diagram: Magazine exchange from the machine to the robot



2.2.5.2 Magazine handover from the robot to the machine

The robot is located here in front of the respective magazine and sends the telegram BOARD_REQUIREMENT when ready to hand over a magazine. The remaining sequences are equivalent to 2.2.5.1.

2.2.5.2.1 Sequence diagram: Magazine exchange from the robot to the machine

2.2.6 Message overview of board/magazine handling

This message signals that a magazine (board available) is available for handover.

BOARD_REQUIREMENT	
<pre><ASYS> <BOARD_REQUIREMENT l_MsgId="1234" s_Type="AES03D" s_SerNum="034567" l_lane="1" l_value="1"/> </ASYS></pre>	
<u>Optional with extension</u> <pre><ASYS> <BOARD_REQUIREMENT l_MsgId="1234" s_Type="AES03D" s_SerNum="034567" l_lane="1" l_value="1" s_MagCode="abcdefg1234" s_MatType="MAT_1234" s_Charge="batch12" s_ProductName="ASYS 001"/> </ASYS></pre>	
l_MsgId	Consecutive number
s_Type	Machine name/type
s_SerNum	Serial number of the machine
l_lane	Lane (1, 2, ...)
l_value	PCB available (1) or not available (0)
l_pass	(0=FAIL, 1= OK), PASS Signal
s_MagCode	Magazine barcode; string(80) (optional, required for material logistics interface)
s_MatType	Material type; string(80) (optional, required for material logistics interface)
s_Charge	Charge; string(80) (optional, required for material logistics interface)
s_ProductName	Product names; string(20) (optional, required for automatic product change)

The attributes marked as optional are switched on/off by configuration parameters on the machine side if the material logistics interface is to be used.

This message signals that a magazine (board permission/release) can be taken over.

BOARD_PERMISSION	
<pre><ASYS> <BOARD_PERMISSION l_MsgId="1234" s_Type="AES03D" s_SerNum="034567" l_lane="1" l_value="1"/> </ASYS></pre>	
l_MsgId	Consecutive number
s_Type	Machine name/type
s_SerNum	Serial number of the machine
l_lane	Lane (1, 2, ...)
l_value	PERMISSION=TRUE (1) or not (0)

This message signals that a magazine (board) has been delivered.

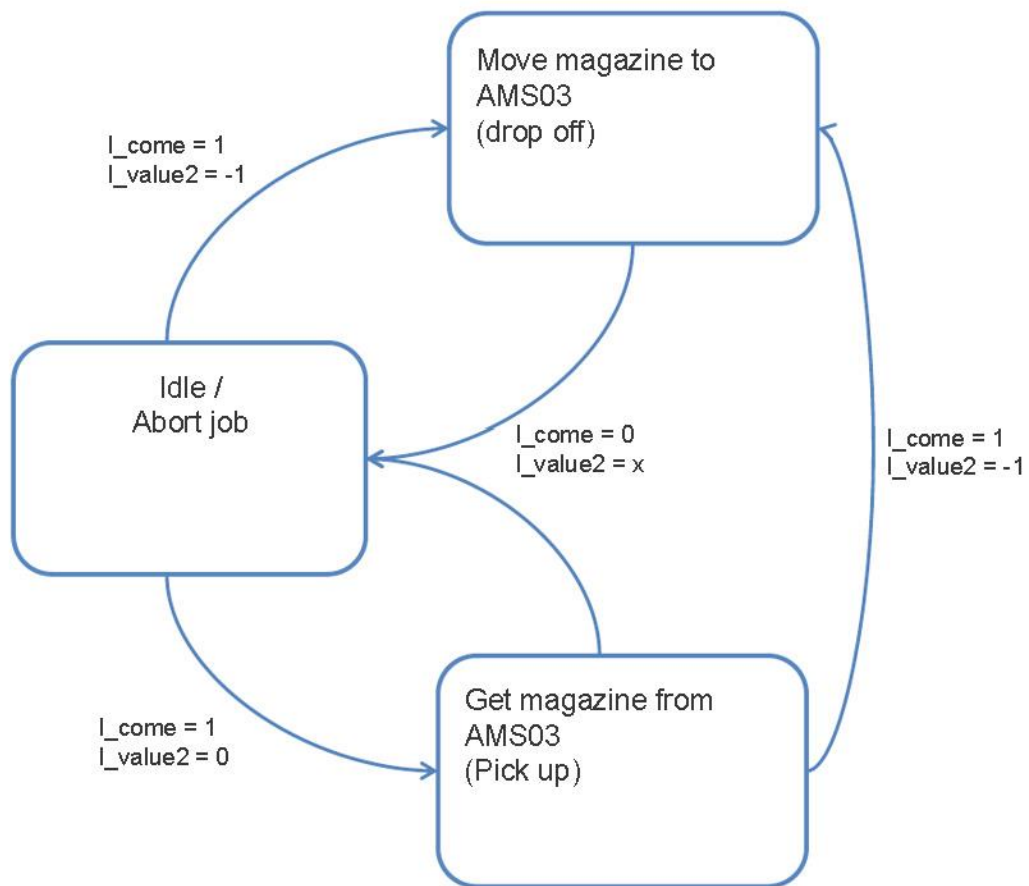
BOARD_DELIVERY	
<pre><ASYS> <BOARD_DELIVERY l_MsgId="1234" s_Type="AES03D" s_SerNum="034567" l_lane="1" l_value="1"/> </ASYS></pre>	
l_MsgId	Consecutive number
s_Type	Machine name/type
s_SerNum	Serial number of the machine
l_lane	Lane (1, 2, ...)
l_value	PCB delivered (1) or not (0)

This message signals that a magazine (board) has arrived.

BOARD_RECEIVAL	
<pre><ASYS> <BOARD_RECEIVAL l_MsgId="1234" s_Type="AES03D" s_SerNum="034567" l_lane="1" l_value="1"/> </ASYS></pre>	
l_MsgId	Consecutive number
s_Type	Machine name/type
s_SerNum	Serial number of the machine
l_lane	Lane (1, 2, ...)
l_value	PCB received (1) or not (0)

3 Appendix

3.1 State diagram of jobs per lane



3.2 Flow chart of magazine transport/warning

